

Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. **8627**

AY: 2025-26	Session: Odd MSE October 2025	Date: 08-10-2025
Sem: 1	Slot: 10:00am - 11:00am	C. Code: CS413
Exam Type: Written	Block: 409	Seat No: 2022700004
Course Section: 1		

L_32_617_986_1_1505_4147



Name of the candidate. Krustubh Bagale

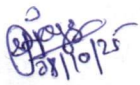
Candidate's UID number:

2 0 2 2 7 0 0 0 0 4

Examination class room number:

4 0 9

Invigilator's signature with date:


08/10/25

(To be filled in by the candidate)

EXAMINATION: Sem VII

(For example FY MCA/FY BTECH)

SEMESTER :

PROGRAM: CSE DS

I / II / III / IV / V / VI / VII / VIII

DAY AND DATE: Wednesday 8/10/25

COURSE NAME:

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting**.
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

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Q1

(b)

Over fitting is the process in which the model learns irrelevant and non-important features of the training data, like noise. And learns ~~the~~ very minute details of the training data set, this results in the model having very high training accuracy but ~~very~~ low accuracy on unseen (validation) data.

When model capacity is too high this means that the model is capable of identifying and/or learning ~~many~~ a wide number of functions and patterns given enough training time. When given a dataset with some noise (outliers) the model learns these features and becomes overfitted.

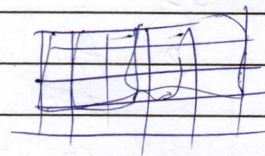
This hinders generalization as the model becomes too specialized according to the ~~the~~ training dataset.

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Q2

(a.)

The output dimensions will be

 6×6

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Q2

(b)

A linear ~~an~~ undercomplete auto encoder is ~~used~~ used for dimensionality reduction. PCA captures the principle components of the data to derive the reduced dimension. They both give the same reduced and compressed dimensions hence the latent space is equivalent to the PCA.

An over complete auto encoder is ~~an~~ one where the dimensions of the latent space is higher than the dimensions of the input.

When this is the case, it is easy for the model to learn the identity function as the dimensions can just be copied ~~to~~ or mapped from the input space to another cell or neuron in the latent space. And the loss function is minimized (without Regularisation) because the decoder can easily map these dimensions back to the output space, to perfectly recreate the input.

This means that the model is not learning any actual insights ~~or~~ or features of the input space but it just learns the identity function and uses it by just copying the input space.

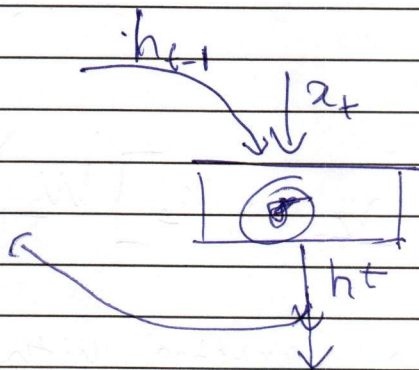
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Q3

(a)

An RNN is a neural network which stores ~~the~~ the output to use in the next call. It is used for sequential data.


~~h_t~~

$$h_t = \sigma(W(x_t, h_{t-1}) + b)$$

An LSTM models - long term and short term memory by keeping a cell-state which is maintained as a long term memory.

While using RNN, the ~~weight update~~ ~~can cause~~ output update can cause an exponential effect due to the weight matrices used. It can cause an exponential increase or decrease ~~in the~~ in the weight update. This causes gradient explosion or gradient vanishing.

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The forget gate of an LSTM Models the cell to "forget" some of the long term memory using information from the short term memory

$$f_t = \sigma([h_{t-1}, x_t] W_f + b_f)$$

this is used in conjunction with the input gate to calculate the new C_t

$$i_t = \sigma([h_{t-1}, x_t] W_i + b_i)$$

$$\tilde{C}_t = \tanh([h_{t-1}, x_t] W_c + b_c)$$

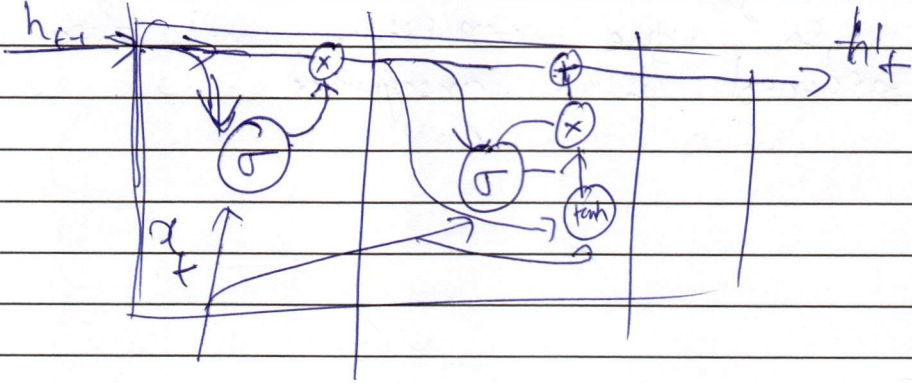
$$C_t = (C_{t-1} \otimes f_t) + (i_t \otimes \tilde{C}_t)$$

This reduces the exponential effect of the weight updates.

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(Q3)



The GRU has two gates the reset gate and the output gate.

~~$$r_t = \sigma([h_{t-1}, x_t] W_r + b_r)$$~~

$$r_t = \sigma([h_{t-1}, x_t] W_r + b_r)$$

~~$$o_t = \sigma([h_{t-1}, x_t] W_o + b_o)$$~~

$$o_t = \sigma([h_{t-1}, x_t] W_o + b_o)$$

$$\tilde{h}_t = \tanh([h_{t-1}, x_t] W_{\tilde{h}} + b_{\tilde{h}})$$

$$h_t = (h_{t-1} \odot r_t) + (o_t \odot \tilde{h}_t)$$

For GRU's, The number of gates is less than that of LSTM. GRU has reset gate and output gate and LSTM has forget gate, input gate and output gate.

GRU have two weight and bias matrices and LSTM will have 3, according to the number of gates in each.



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This will lead to generalization as the capacity of the model will be decreased. So the overfitting will also be reduced as a consequence.

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Sr. No. 5838

8627



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